MEDASSISTAI - DISEASE PREDICTION SYSTEM

BACKEND DEVELOPMENT REPORT (MILESTONE 1)

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TEAM - 2

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1. PROJECT OVERVIEW

Introduction

MedAssistAI is an AI-powered healthcare platform designed to assist patients and doctors through a secure digital environment. The backend of the application is developed using **FastAPI**, while **PostgreSQL** is used as the relational database. During Milestone 1, the primary objective was to establish a secure backend foundation by implementing authentication and authorization mechanisms.

My Contribution

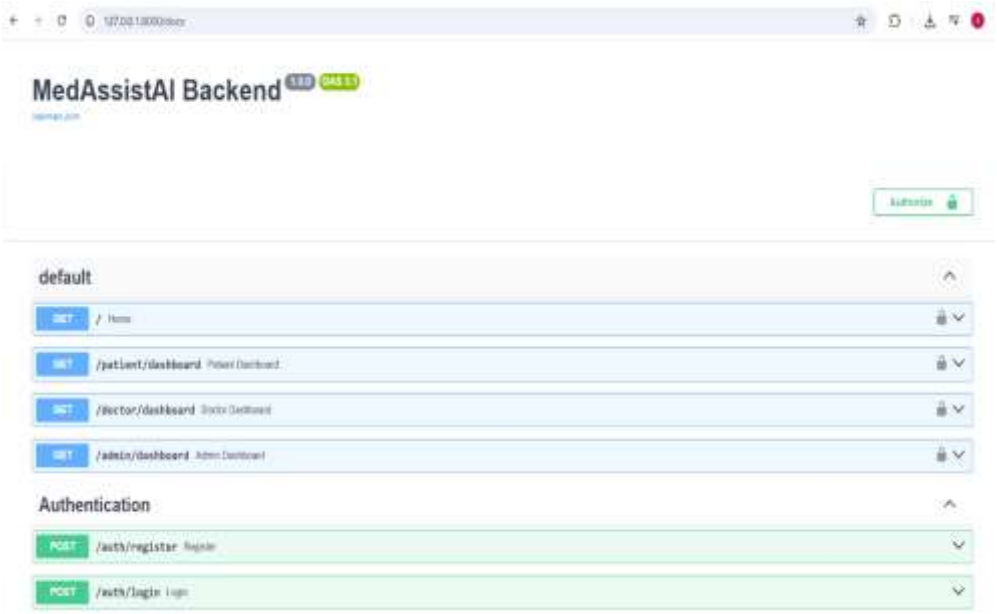
As the backend developer, my responsibilities included:

- Configuring the FastAPI backend.
- Integrating PostgreSQL using SQLAlchemy.
- Developing user registration and login APIs.
- Implementing password hashing using bcrypt.
- Generating JWT tokens for authentication.
- Protecting APIs using JWT verification.
- Implementing Role-Based Authorization for different user roles.

2. OBJECTIVES

The objectives of Milestone 1 were:

- Develop the backend using FastAPI.
- Configure PostgreSQL database connectivity.
- Implement secure user registration.
- Implement secure user login.
- Encrypt passwords before storing them.
- Generate JWT authentication tokens.
- Protect backend APIs.
- Restrict dashboard access using user roles.



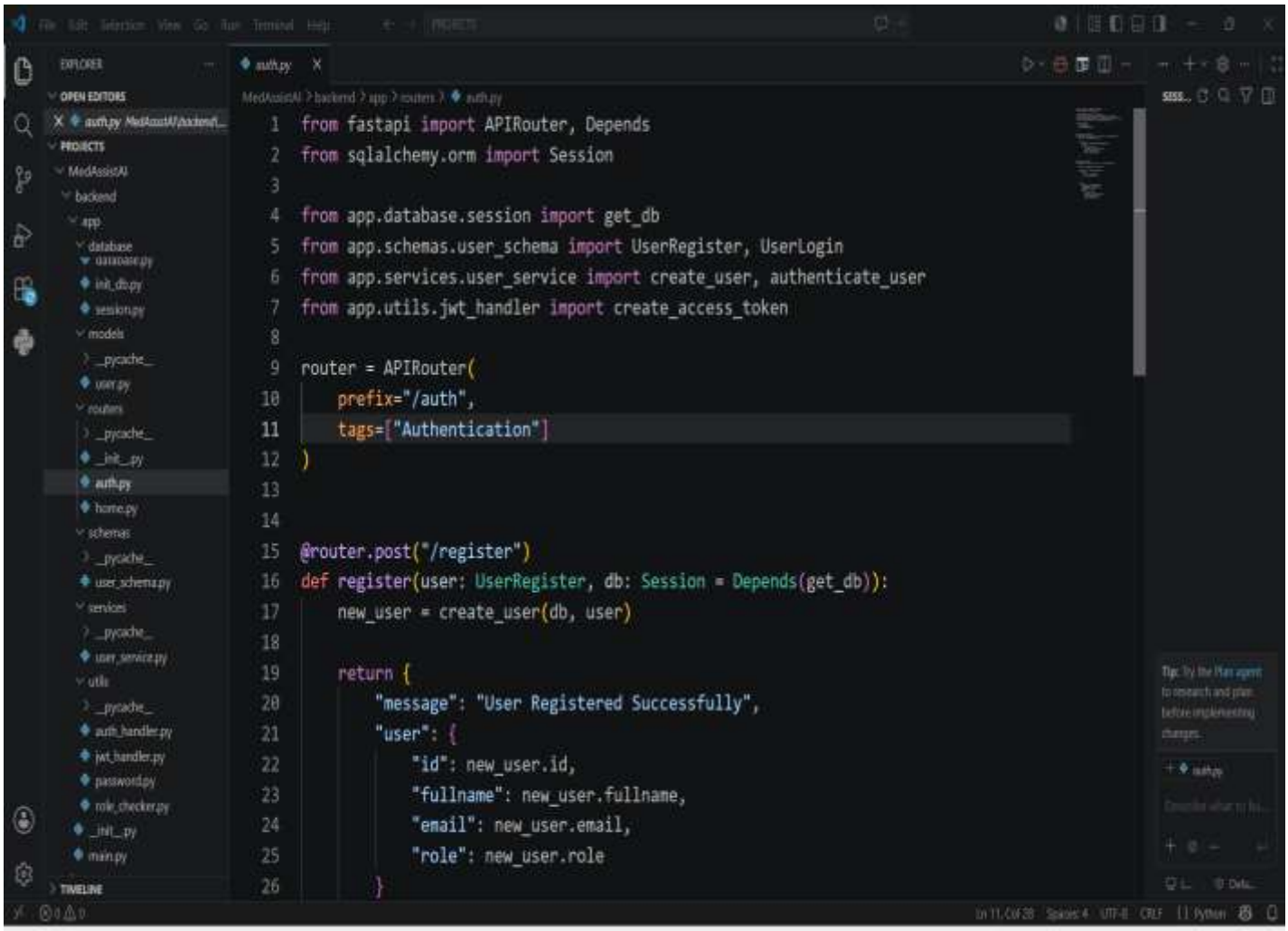
3. TECHNOLOGIES USED

Why These Technologies?

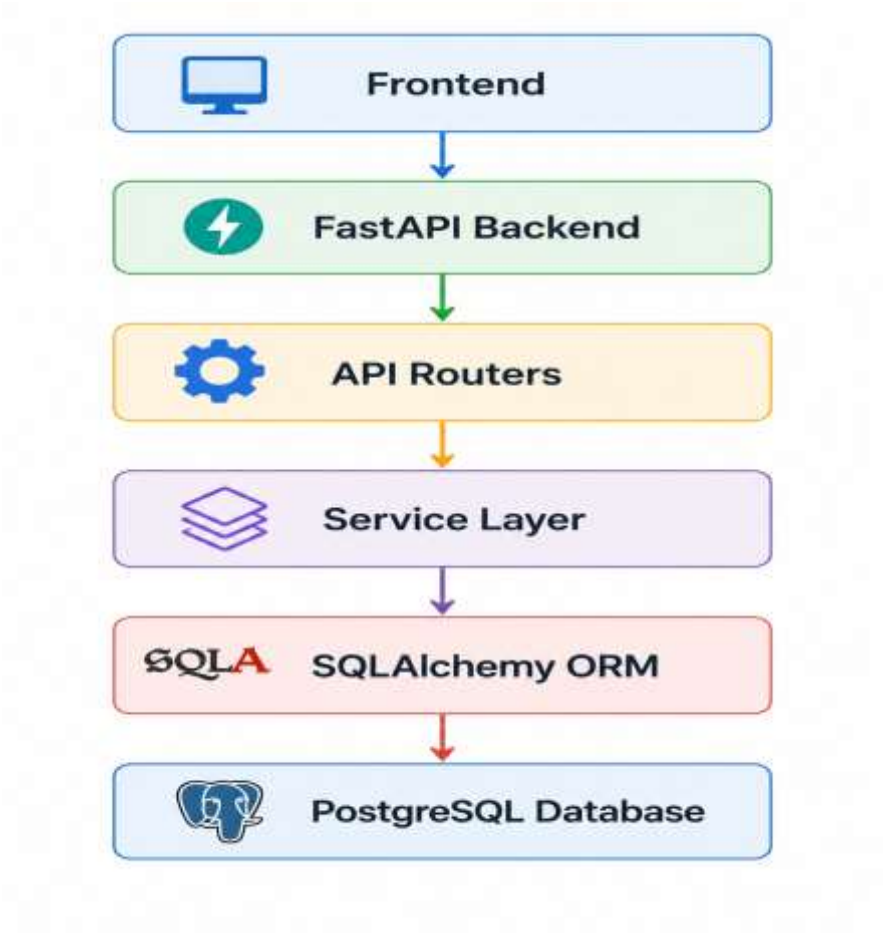
- FastAPI provides high-performance backend development.
- PostgreSQL ensures secure and reliable data storage.
- SQLAlchemy simplifies database operations.
- JWT provides secure authentication.
- Swagger enables easy API testing.

Technology	Purpose
Python 3.13	Programming Language
FastAPI	Backend Framework
PostgreSQL	Relational Database
SQLAlchemy	ORM
Pydantic	Data Validation
bcrypt	Password Hashing
python-jose	JWT Authentication
Swagger UI	API Testing
Git & GitHub	Version Control

4. BACKEND PROJECT STRUCTURE



5. BACKEND ARCHITECTURE



Explanation

- The frontend sends API requests.
- FastAPI receives the request.
- Routers identify the appropriate endpoint.
- Services execute business logic.
- SQLAlchemy communicates with PostgreSQL.
- The response is returned to the frontend.

6. DATABASE CONFIGURATION

The project uses **PostgreSQL** as the relational database.

Why PostgreSQL?

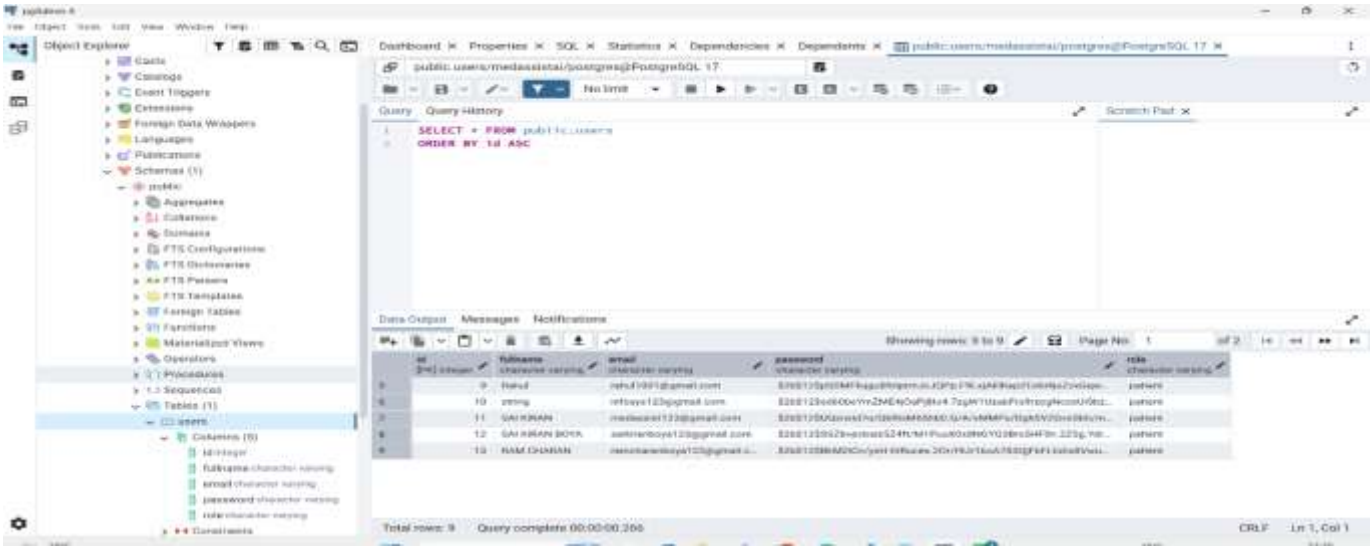
- Open Source
- Secure
- Reliable
- High Performance
- Supports Large Data

Database Configuration

Database connection is established using SQLAlchemy

User information stored includes:

- User ID
- Full Name
- Email
- Password (Hashed)
- Role



7. AUTHENTICATION MODULE

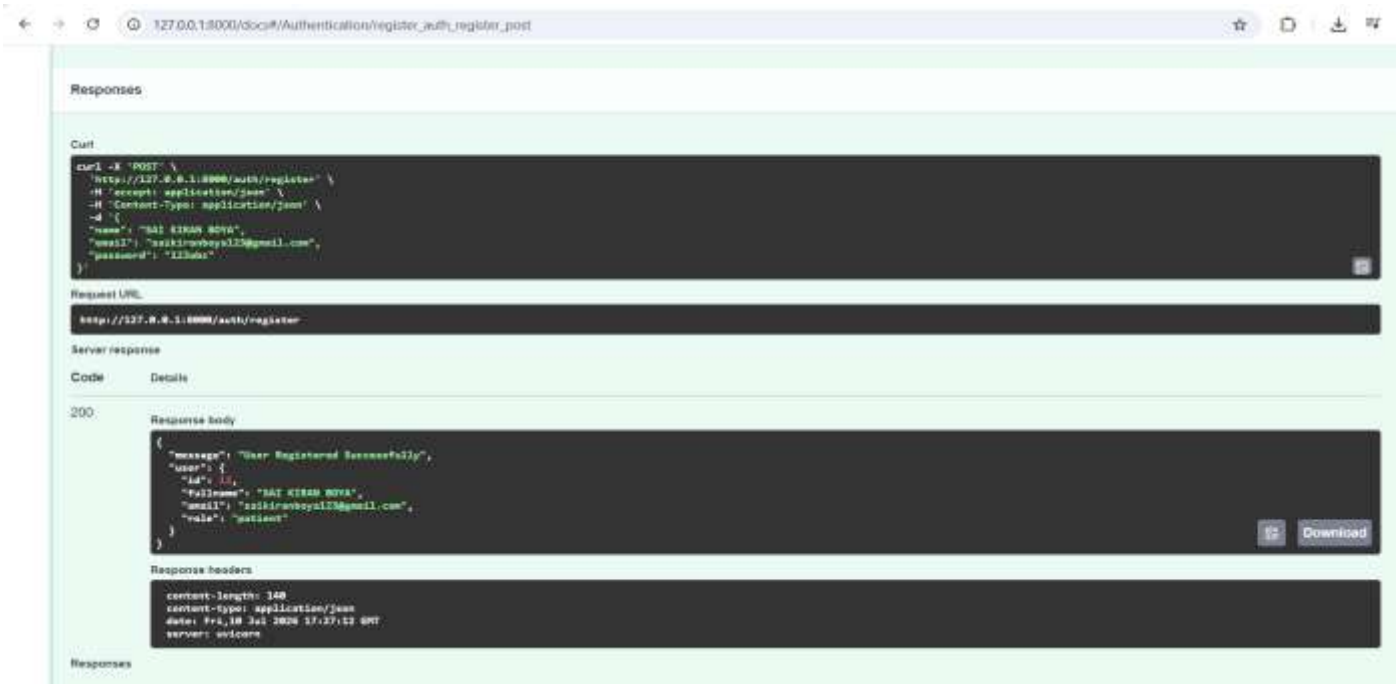
Authentication Module
Authentication ensures that only valid users can access the system.
Two APIs are implemented.

Register API
The user provides:

- Full Name
- Email
- Password

The backend performs:

- Input validation
- Duplicate email validation
- Password hashing
- Stores data in PostgreSQL



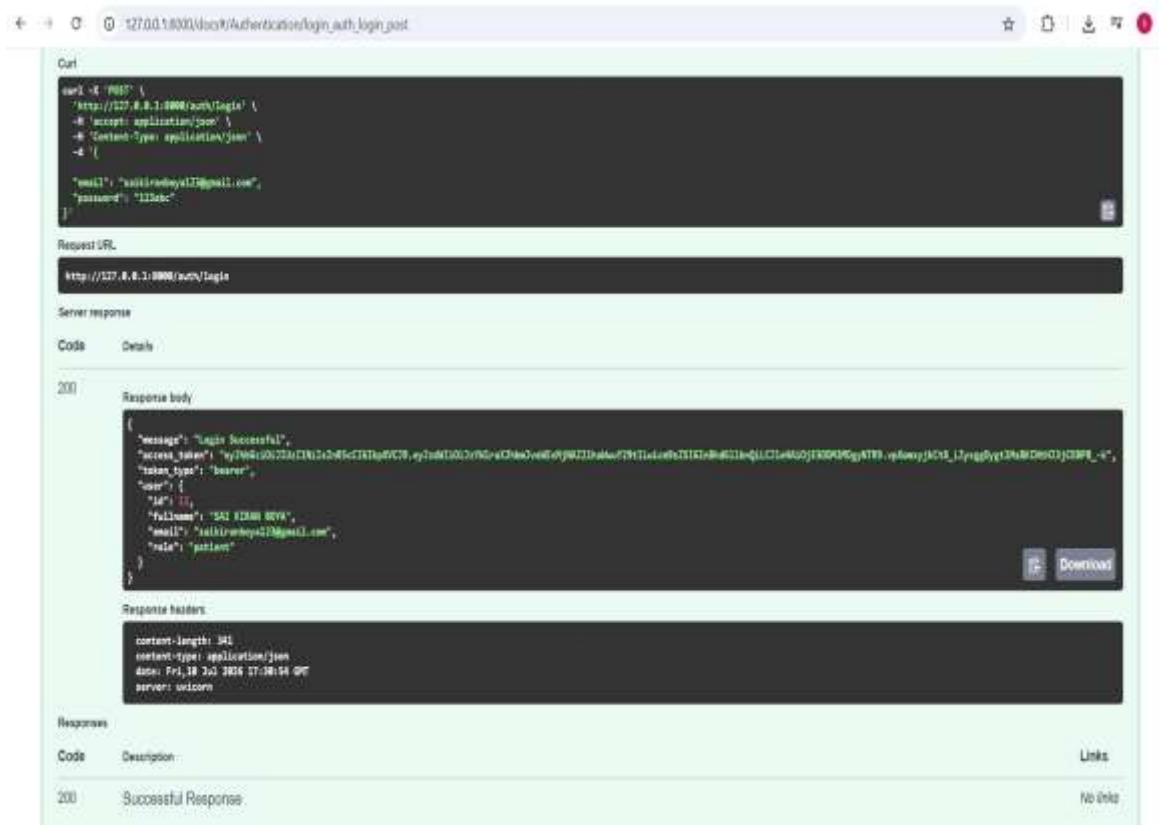
LOGIN API

The user enters:

- Email
- Password

The backend:

- Verifies credentials
- Checks hashed password
- Generates JWT Access Token
- Returns Login Success response

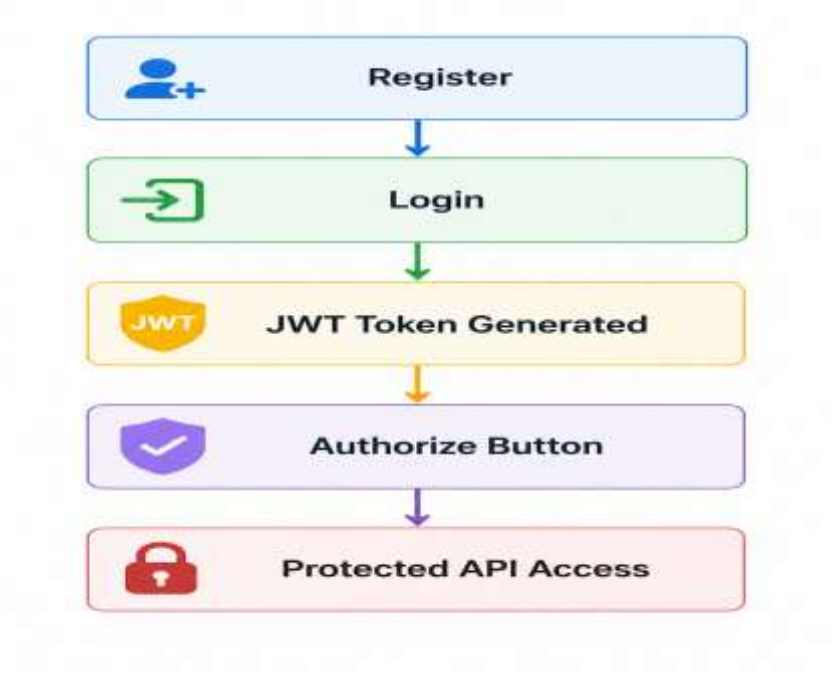


8. JWT AUTHENTICATION

JWT contains

- User Email
- User Role
- Expiration Time

AUTHENTICATION FLOW



9. ROLE-BASED AUTHORIZATION

Different users have different permissions.

Patient

Can access
/patient/dashboard

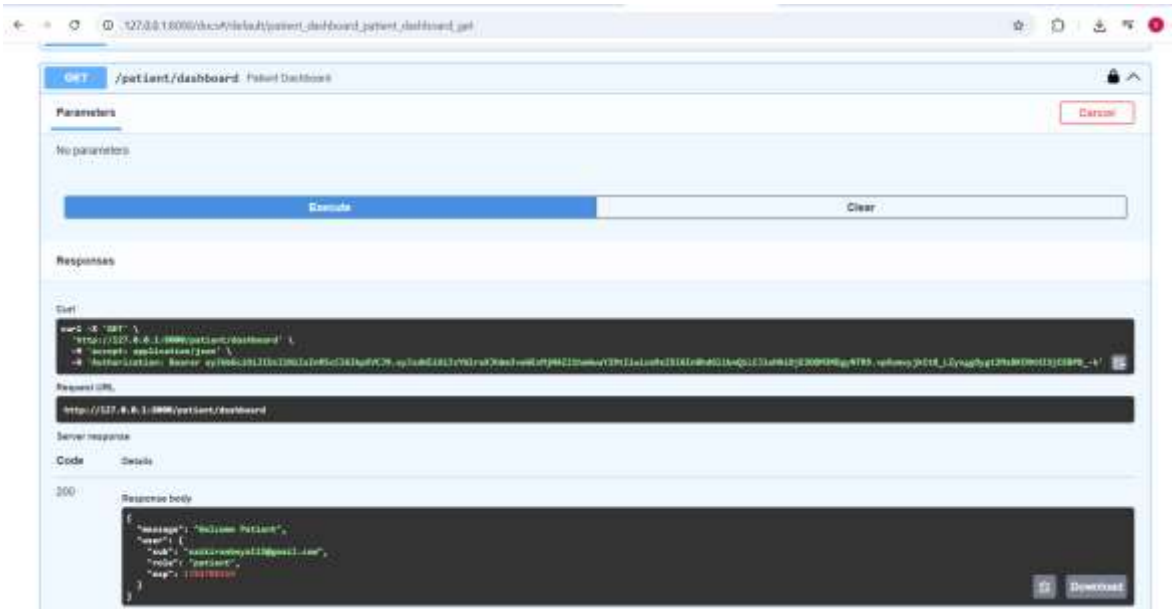
Doctor

Can access
/doctor/dashboard

Admin

Can access
/admin/dashboard

If a Patient tries to access Doctor or Admin Dashboard,
Backend returns
HTTP 403 Forbidden
This prevents unauthorized access.



10. API TESTING

SUCCESSFULLY TESTED APIS	
METHOD	ENDPOINT
POST	/auth/register
POST	/auth/login
GET	/patient/dashboard
GET	/doctor/dashboard
GET	/admin/dashboard

Testing Results:

- Registration Successful
- Login Successful
- JWT Generated Successfully
- Patient Dashboard Accessible
- Unauthorized Access Returns HTTP 403

11. CONCLUSION

Milestone 1 successfully established the backend foundation of the MedAssistAI project.

Achievements:

- FastAPI Backend Developed
- PostgreSQL Database Integrated
- SQLAlchemy ORM Configured
- User Registration Implemented
- User Login Implemented
- JWT Authentication Implemented
- Role-Based Authorization Implemented
- APIs Successfully Tested

Future Scope:

- In the upcoming milestones, the backend will be extended by integrating:
- Patient Module
- Doctor Module
- Appointment Module
- Medical Reports Module
- AI Healthcare Features
- Frontend Integration